PCRQ 11

Graded

Student

VK Nguyen

Total Points

3 / 4 pts

Question 1

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 2

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 3

(no title) **0** / 1 pt

+ 1 pt Correct

✔ **+ 0 pts** Incorrect

Question 4

(no title) **1** / 1 pt

✔ **+ 1 pt** good faith answer

+ 0 pts Blank

this is what we prove on pg 7, it needs some work to prove using the sin of sum and cos of sum of angles identities!

Q1
1 Point

Suppose that $|z| = 2, |w| = 3$, and that arguments of z and w are $\frac{\pi}{6}$ and $\frac{\pi}{4}$ respectively. Which statement about zw is correct?

- $|zw| = 5, \arg(zw) = \frac{5\pi}{12}$
- $|zw| = 6, \arg(zw) = \frac{\pi}{12}$
- **$|zw| = 6, \arg(zw) = \frac{5\pi}{12}$.**
- $|zw| = 6, \arg(zw) = \frac{\pi}{24}$.

Q2
1 Point

Which statement correctly describes the n-th roots of unity?

- They all lie on the real axis.
- They all have modulus n.
- **They are equally spaced around the unit circle.**
- There is only one n-th root of unity.

Q3
1 Point

Suppose $z, w \in \mathbb{C}$ and $|z| = 3$ and $|z + w| = 7$. Which of the following must be true?

- **$|w| \leq 4$.**
- $|w| \geq 4$.
- $|w| = 4$.
- $|w| \geq 10$.

Q4

1 Point

Please write briefly (in two or three sentences) what single point of the notes of Lecture 12 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

I have a question regarding polar form. Why does in polar form multiplying complex numbers mean multiplying their lengths and adding their angles?